

A Transfer Learning Method based on VGG-16 Convolutional Neural Network for MI Classification

Ming-ai Li ^{1,2,3}, Dong-qin Xu ¹

1. Faculty of Information Technology, Beijing University of Technology, Beijing 100124, China

2. Beijing Key Laboratory of Computational Intelligence and Intelligent System, Beijing 100124, China

3. Engineering Research Center of Digital Community, Ministry of Education, Beijing 100124, China

E-mail: limingai@bjut.edu.cn

E-mail: xudq@emails.bjut.edu.cn

Abstract: Brain computer interface (BCI) technology can help the disabled to achieve the recovery of neural function by using the Motor Imagery Electroencephalogram (MI-EEG) based rehabilitation system. However, it is difficult to acquire a large amount of available EEG data, transfer learning technology provides an effective method, and the source domain selection is one of key issues. In this study, we develop a novel parameter transfer learning method based on VGG-16 convolutional neural network (CNN) for MI classification. First, the number of all MI-EEG signals are augmented with the sliding window method, and the short-time Fourier transformation (STFT) is applied to obtain the time-frequency spectrum images (TFSI). Then, the VGG-16 CNN is pre-trained with TFSI of source domain, which is divided into five blocks. The parameters of the pre-trained CNN are transferred to the target network through a new transfer strategy, i.e. utilization of the data of part subjects from target domain to fine-tune the five blocks in turn. Finally, the fine-tuned CNN is used for MI classification of the rest subjects in target domain. This work is evaluated with a public dataset, the best classification accuracy of this study is 96.59%. The results show that the high correlation between the source domain and the target domain is better than using the domains with low correlation, and the proposed transfer strategy is efficiency.

Key Words: Brain computer interface, Motor Imagery, Transfer learning, Convolutional neural network, Transfer strategy

1 INTRODUCTION

Brain computer interface (BCI) technology can control external devices through cognitive activities, the disabled who need rehabilitation may achieve the recovery of neural control system through continually training [1-2]. The Electroencephalography (EEG) uses voltage fluctuations on the scalp to show brain activity, especially during Motor Imagery (MI) [3-4]. However, it is expensive to acquire a large amount of data and labeling in BCI, and new subjects need to undergo a lot of training to adapt the EEG acquisition equipment, these will consume considerable time. In recent years, the transfer learning technology can transfer the learned knowledge from source domain to rapidly adapt the target domain, and it is significant for BCI. At present, the transfer learning has made great progress in fields of image recognition, bioelectrical signal recognition and so on, it is considered to be the important research direction of MI-EEG classification [5-7]. The transfer learning approaches in BCI can be summarized into three contexts. The first context is the instance transfer learning approach [8-10], which assumes that certain data from the source domain can be transferred to the target domain and reused. The second context is the feature transfer learning approach [11-13], the common feature representation of the source and target domain is identified, then these features are used to transfer knowledge. The last context is the

parameter transfer learning approach [14,15], the main aim is to share model parameters or prior knowledge between the source and target domains. Due to the practicality and easy realization of the parameter transfer learning method, which has been more popular in EEG analysis [16-21]. Hajinoroozi et al. [16] conducted a comprehensive research on the transferability of the deep convolutional neural network (CNN) for cross-experiment and cross-subject prediction based on rapid serial visual presentation events. Wu et al. [17] proposed a parallel multiscale filter bank CNN (MSFBCNN), which is applied to extract temporal and spatial features, to classify the MI-EEG signals. Recently, an end-to-end deep CNN model is developed to get rid of the hidden focus of EEG event-related potentials during object selection [18-19]. To mitigate the impact of cross-dataset variability, an online pre-alignment strategy was proposed to adjust the EEG distribution of different subjects by Xu et al. [20]. Zhang et al. [21] proposed a hybrid deep neural network framework with transfer learning to decode and recognize the temporal and spatial features of MI signals. Gaowei et al. [22] proposed a deep transfer neural network framework by freezing the parameters of the front 11 layers, which were transferred from a pre-trained VGG-16 CNN based on ImageNet, and the parameters of the rest layer are fine-tuned. However, whether the knowledge of source domain is suitable to be transferred in BCI is not fully considered. In addition, it is also worth studying that how to use part data of the same dataset to assist new subjects in classification.

This work is supported by National Nature Science Foundation under Grant No. 11832003 and No. 81471770.

In this paper, we study the parameter transfer learning for different subjects in the different/same dataset and propose a novel parameter transfer strategy based on VGG-16 CNN for MI classification. First, raw MI-EEG signals of C3, C4 and Cz electrodes from source domain are selected, the

where the size of all the convolution kernels is 3×3 , the number of convolution kernels is 64, 128, 256, 512 and 512 with the corresponding number of convolution layers is 2, 2, 3, 3 and 3 respectively, and the activation functions are set as Rectified Linear Unit (ReLU). The 5 pooling layers is

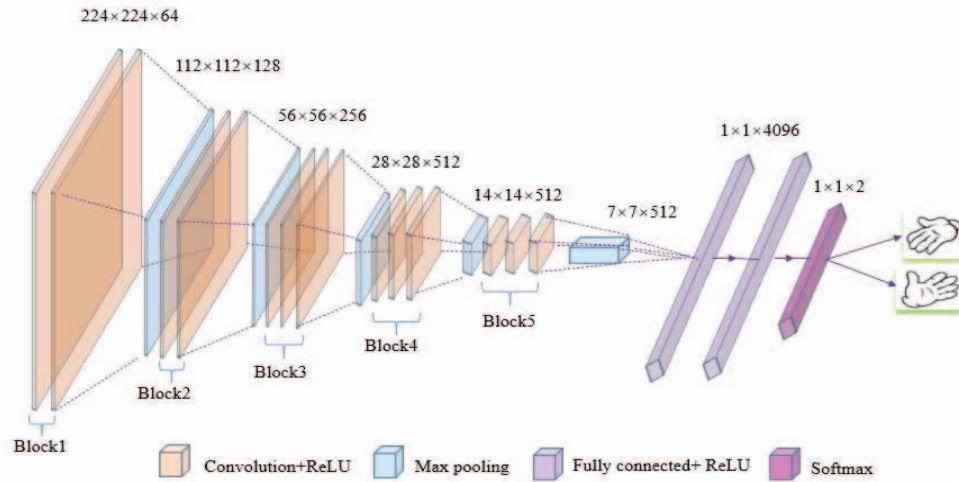


Fig 1. The structure of improved VGG-16 convolutional neural network.

short-time Fourier transformation (STFT) is used to extract the time-frequency features of EEG signals and obtain the time-frequency spectrum images (TFSI). After, a VGG-16 CNN model is pre-trained on the TFSI of source domain, the parameters of the pre-trained model are transferred to the target model. Then, the target model is fine-tuned with part subjects of the target domain, and which is applied to classify the MI-tasks of rest subjects. This work suggests that the more correlation between two different domains, the more beneficial to transfer learning in MI-EEG, and the proposed parameter transfer learning method can reduce individual training time.

This paper is organized into 5 sections. In Section 2, the proposed parameter transfer learning method is introduced. Experimental research is performed on a publicly available dataset in section 3. Section 4 makes results and the conclusions are drawn in the final section.

2 TRANSFER LEARNING METHOD

In this section, we introduce a novel parameter transfer learning method based on VGG-16 CNN, which is mainly divided into following 3 steps.

Step 1: Design of network model. VGG-16 is a deep convolutional neural network developed by researchers from the Visual Geometry Group of Oxford University and Google DeepMind, which is a very powerful classification model, especially in various image [23]. The output layer of original VGG-16 network is replaced with a new output layer, where the number of neurons equals to that of classes of target MI tasks as well as the softmax classifier is applied for MI-EEG classification. Then, the layers with convolution structure are divided into 5 blocks depending on whether the number of convolution kernels is equal. The detailed structure of VGG-16 convolutional neural network and the 5 blocks are demonstrated in Figure 1 and Table 1.

Step 2: Preparation of data. In order to obtain time-domain and frequency-domain features of raw MI-EEG at the same time, STFT is applied to obtain the TFSI. In addition, the obtained TFSI need further processing because the input shape of VGG-16 is $224 \times 224 \times 3$, where 224×224 is the

Table 1. Detailed architecture of the improved network

	Layer	Kernel	Number of Kernel	Number of Layer
	Input	Size= $224 \times 224 \times 3$		
Block1	Convolution	(3×3) Stride=1	64	2
	Max Pooling	2×2 Stride=2	1	1
Block2	Convolution	(3×3) Stride=1	128	2
	Max Pooling	2×2 Stride=2	1	1
Block3	Convolution	(3×3) Stride=1	256	3
	Max Pooling	2×2 Stride=2	1	1
Block4	Convolution	(3×3) Stride=1	512	3
	Max Pooling	2×2 Stride=2	1	1
Block5	Convolution	(3×3) Stride=1	512	3
	Max Pooling	2×2 Stride=2	1	1
	Fully Connected	Dense out=4096		2
	Output	Dense out=2		

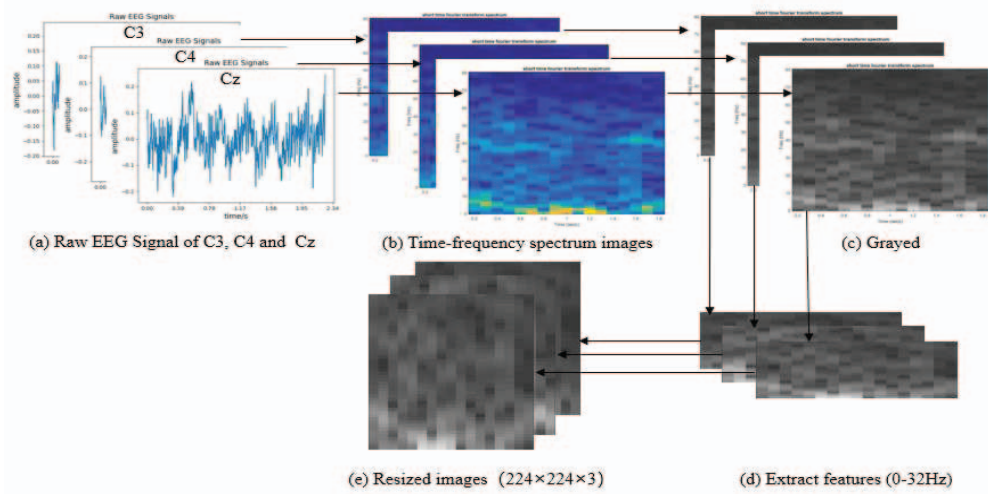


Fig 2. The conversion process from raw MI-EEG signals to TFSI.

two-dimensional size of the image, and 3 represents layers of color with red, green, and blue (RGB) intensity for a colored image. For source domain, 3 images of 0-32 Hz frequency bands are extracted from channels C3, C4 and Cz as one trails. Besides, to match the input shape of VGG-16 CNN, the all images are grayed and resized to 224×224 . Then, these $224 \times 224 \times 3$ images are fed into the VGG-16 CNN for pre-training. The detailed conversion process of raw MI-EEG signals to TFSI is displayed in Figure 2. The raw EEG signals of target domain has the same process as well.

Step 3: Improvement of transfer strategy. To obtain a better performance of transfer learning in this study, a novel transfer strategy is proposed to optimize the transferred parameters. First, the parameters of pre-trained network are transferred to the target network with the same network configurations as pre-train network. Then, for different subjects in the same target dataset, there are i subjects who were randomly selected from the dataset to fine-tune the parameters of transferred blocks in turn, where $i = 1, 2, \dots, 4$. For example, the block 1 is frozen, the subject

1 is used to fine-tune the parameters of rest blocks. After, the block 1 and block 2 are frozen, the subject 2 is used to fine-tune the parameters of block 3, block 4 and block 5, where the frozen block 2 is fine-tuned by 1st subject. By that analogy, the last block is fine-tuned by 4 subjects. Finally, each fine-tuned network will be used for MI classification of the rest subjects (as the new subjects) in the same dataset. The fine-tune process of 4 subjects and the proposed transfer strategy is shown in Figure 3.

3 EXPERIMENT

In this study, PhysioNet EEG Motor Movement/Imagery Dataset [24] and Dataset 2b from BCI Competition IV [25] are used to evaluate the performance of the proposed parameter transfer learning method. In addition, a sliding window method is exploited for augmenting the data. Then, the obtained TFSI of source domain are used to train the pre-trained network model, and the fine-tuned model is applied to classify MI-EEG of different subjects in the target domain.

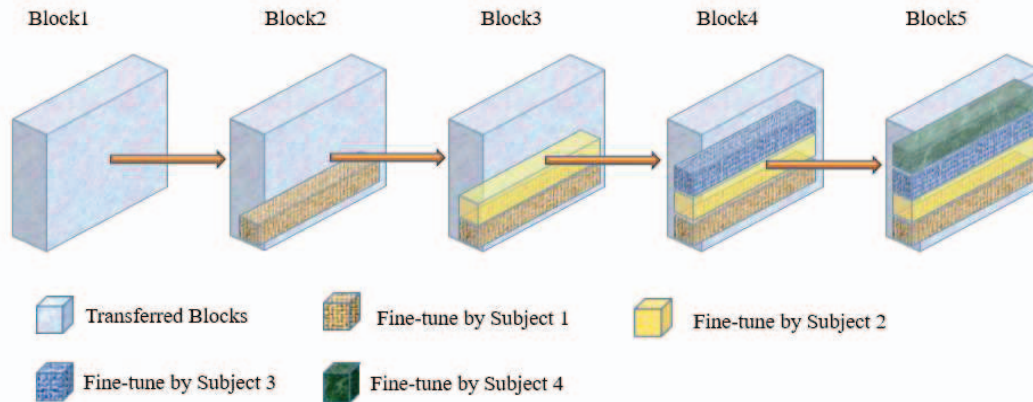


Fig 3. The fine-tune process of proposed transfer strategy.

3.1 Dataset Description

Source domain: The left and right hands MI data of 98 subjects with the same timestamp alignments (45 4s trials per subject) are chosen from the PhysioNet EEG Motor Movement/Imagery Dataset, which consisted of 109 subjects, as the source domain to pre-train the network model. The MI-EEG were recorded by BCI2000 instrumentation system with a sampling frequency of 160 Hz in 64 electrodes.

Target domain: The first 4 subjects of Dataset 2b from BCI competition IV is applied to fine-tune the parameters of transferred blocks in turn, and rest 5 subjects as the target subjects. The MI-EEG were obtained from C3, Cz and C4 three electrodes at a sampling frequency of 250 Hz. This dataset consists of three training sessions and two test sessions (120 trials per session) of left and right hands MI. Here, the three sessions of training were used.

3.2 Dataset Processing

In this experiment, the original data was preprocessed as follows. Firstly, to make the source domain and target domain more correlation, here acquired data by C3, C4 and Cz electrodes, which has the same position with target domain, are selected from 64 electrodes of the source domain. Figure 4 displays the electrode positions.

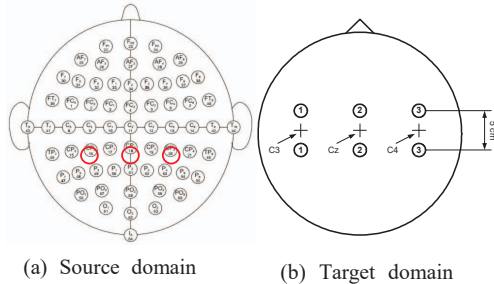


Fig 4. Electrode positions.

Secondly, we augment the data because training a deep neural network requires a large available data. For source domain, the total trails are increased 21 times by using the sliding window (2 seconds) with a step of 0.1s. The data augment process is shown in Figure 5. For target domain, the duration between 3.5s and 6.5s MI tasks of each EEG trail is used and the total trails are increased 11 times using the same method with source domain.

Finally, in order to evaluate the performance of our method, extensive experiments are conducted based on two public datasets. We first used 92610 trails from source domain to train the improved VGG-16 CNN, and all parameters of the pre-trained model were transferred to target network model. After, the block 1 was frozen, and the TFSI of subject 1 from target domain were used to fine tune the transferred parameters of rest blocks, then the fine-tuned target model was applied to classification on subject 5, 6, 7, 8 and 9. The experiment process of subject 2, 3 and 4 for fine-tuning followed the same as that of subject 1, and 5 results for each subject, which is showed in Table 2. Moreover, a pre-trained network based on ImageNet was also frozen in

turn, but the proposed transfer strategy was not applied, in order to compare with our method, the last 5 subjects from target domain were selected here and the results were demonstrated in Table 4.

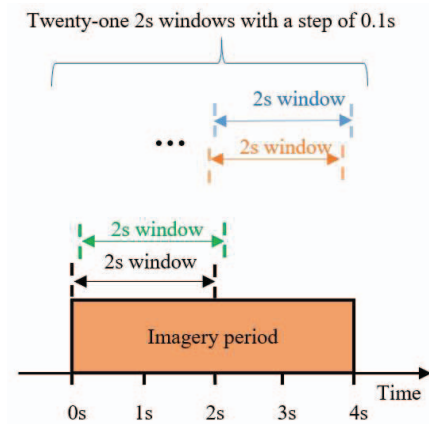
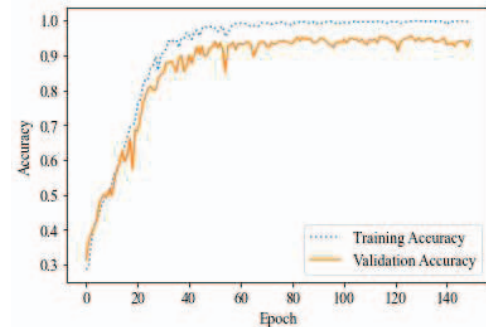


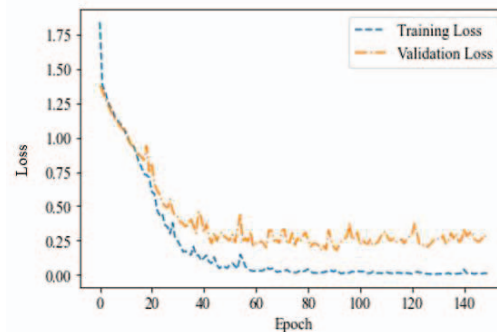
Fig 5. The process of data augment.

4 RESULTS

In this section, we used classification accuracy to evaluate the performance of our method, and yielding a relatively higher classification result. The training progress of pre-trained VGG-16 CNN model is displayed in Figure 6, where the accuracy and loss are showed in (a) and (b)



(a) Pre-trained model accuracy.



(b) Pre-trained model loss.

Fig 6. Training progress plots of pre-trained network model.

respectively. The results indicated that the pre-trained model have learned the knowledge of the source domain. All the experiments are carried out at the same software (Spyder, Windows10) and the hardware (a Hewlett-Packard computer, which is equipped with an Inter(R) Core (TM) i7-9700 CPU @ 3.00GHz, a NVIDIA GeForce RTX 2070 GPU) environment.

To verify the performance of the proposed parameter transfer learning method in this work, the classification accuracy of our method are compared with [22] and the results are showed in Table 2. Here FB1 presents the first block is frozen, FB1-2 presents the first and second blocks are frozen, FB1-3, FB1-4 and FB1-5 analogize in turn. The results demonstrated that our method outperform the proposed method in [22], and the best performance of this study is 96.59%, 18.49% superior than [22] on subject 8. Besides, it can be obtained from the fourth row of Table 2 that the greatest performance difference between our method and [22] is 23.75% on subject 6. These results all displayed the effectiveness of the proposed method. From Table 3, it can be seen that training the proposed transfer CNN framework for 5 subjects is much faster than the s-CNN, which verified that the proposed method can reduce individual training time. Where, the s-CNN is trained from scratch and the structure of it is shallower than the standard VGG-16 CNN model. In addition, the results also be evaluated by using the 10-fold cross-validation (CV) and show in Figure 7, and the average classification rate with 10-fold CV of 5 subjects is 92.96%, which is higher than [22] (82.42%). Where, S5, S6, S7, S8 and S9 presents Subject5, Subject6, Subject7, Subject8, and Subject9, respectively.

Table 2. Classification results of the proposed method and [22]

	FB1	FB1-2	FB1-3	FB1-4	FB1-5	[22]
Subjects	Accuracy (%)					
5	94.81	93.83	95.48	94.59	93.86	81.30
6	93.30	93.63	92.95	93.64	94.55	70.80
7	94.43	95.57	96.25	92.84	96.14	73.30
8	94.73	96.59	95.25	94.21	95.14	78.10
9	94.89	96.14	95.91	92.61	95.00	76.40

Table 3. Training time of the proposed method and the s-CNN

	FB1-3	s-CNN
Subjects	Training time (s)	
5	453.95	1267.99
6	423.63	1017.89
7	430.73	1338.60
8	514.99	1618.40
9	555.15	1305.64

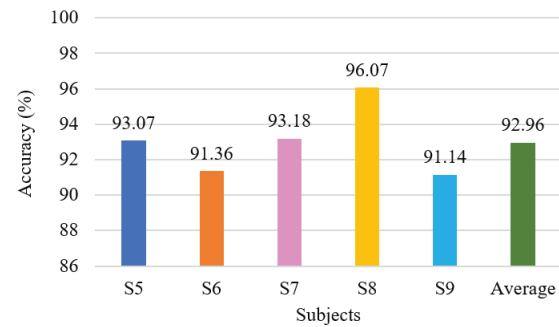


Fig 7. The classification accuracies of 10-fold CV.

Table 4. Classification results of the pre-trained original VGG-16 CNN based on ImageNet and the standard CNN [22]

	I-B1	I-B1-2	I-B1-3	I-B1-4	I-B1-5	CNN [22]
Subjects	Accuracy (%)					
5	68.83	88.65	82.30	75.87	64.61	79.60
6	66.02	73.98	72.50	60.57	56.48	69.20
7	70.00	78.39	75.70	60.45	65.23	69.20
8	76.21	81.02	79.05	72.52	75.83	71.90
9	72.23	75.91	74.32	67.84	48.41	72.60

In order to validate the importance of correlation between the source domain and the target domain for transfer learning in MI-EEG, the classification accuracy of the standard CNN and the VGG-16 CNN model which is pre-trained based on ImageNet are compared with the proposed method in this paper. The classification results of the comparative methods are showed in Table 4. It's similar with Table 2, I-B1, ..., I-B1-5 indicates the number of frozen blocks, the difference is that the proposed transfer strategy is not used to fine tune the transferred parameters, i.e. the parameters of block 2, 3, 4 and 5 is not fine-tuned by subject 1, 2, 3 and 4 in turn. As can be found from the Table 2 and Table 4 that no matter which blocks are frozen, EEG-based pre-trained model is superior than ImageNet-based, which fully verifies the key of source domain selection. Furthermore, comparing the last 2 columns in Table 4, it can be seen that if all the learned knowledge from the source domain is transferred to the target domain, the effect is worse than that without transfer learning technology, indicating that negative transfer occurs, which is due to the low correlation between the source domain and the target domain.

5 CONCLUSIONS

Aiming at using existed knowledge to assist new subjects for MI tasks classification, a VGG-16 CNN based parameter transfer learning method is developed to reduce the heavy demand for training data in this paper. Using sliding window method to augment the number of training data, the obtained TFSI of source domain by STFT is applied to pre-train the VGG-16 CNN model, and the layers with convolution is divided into five blocks. Then, the pre-trained model parameters are transferred to the target

network via the proposed transfer strategy, and the fine-tuned network is used for MI classification. The results show that the pre-trained network in this work was superior than ImageNet-based, it revealed that the more correlation between the source domain and the target domain will perform better classification results than less. In addition, the proposed method suggests that the fine-tuned parameters by the subjects were used for classification in the same dataset outperform no fine-tuned. In the future, we will focus on using the transfer learning to achieve incremental learning and few-shot learning, which can improve the ability of neural networks in the new tasks by retaining the learned knowledge from the old tasks, and reduce the demand for new training data. It provides a new method for rehabilitation engineering research.

REFERENCES

- [1] G. Schalk, D.J. McFarland, T. Hinterberger, N. Birbaumer, and J.R. Wolpaw. BCI2000: A General-Purpose Brain-Computer Interface (BCI) System. *IEEE Transactions on Biomedical Engineering*, Vol.51, No.6, 1034–1043, 2004.
- [2] M. Sebastián-Romagosa, Cho W, Ortnier R, et al. Brain Computer Interface Treatment for Motor Rehabilitation of Upper Extremity of Stroke Patients—A Feasibility Study. *Frontiers in Neuroscience*, Vol.14, No. 591435, 2020.
- [3] M. H. Alomari, A. Samaha and K. Alkamha. Automated Classification of L/R Hand Movement EEG Signals using Advanced Feature Extraction and Machine Learning. *International Journal of Advanced Computer Science and Applications*, Vol.4, No.6, 207-212, 2013.
- [4] Nicolas-Alonso, Luis Fernando, Gomez-Gil and Jaime. Brain computer interfaces: a review. *Sensors*, Vol.12, No.2, 1211-1279, 2012.
- [5] S. J. Pan and Q. Yang. A Survey on Transfer Learning. *IEEE Transactions on Knowledge and Data Engineering*, Vol.22, No.10, 1345-1359, 2010.
- [6] A. Argyriou, A. Maurer and M. Pontil. An Algorithm for Transfer Learning in a Heterogeneous Environment. *Proc. European Conf. Machine Learning and Knowledge Discovery in Databases*. Antwerp, Belgium, 2008.
- [7] Simanto Saha and Mathias Baumert. Intra- and Inter-subject Variability in EEG-Based Sensorimotor Brain Computer Interface: A Review. *Frontiers in Computational Neuroscience*, Vol.13, No.87, 2020.
- [8] I. Hossain, A. Khosravi and S. Nahavandhi. Active transfer learning and selective instance transfer with active learning for motor imagery based BCI. 2016 International Joint Conference on Neural Networks (IJCNN). Vancouver, BC, Canada, 2016.
- [9] H. Ibrahim, K. Abbas, H. Imali, et al. Multiclass Informative Instance Transfer Learning Framework for Motor Imagery-Based Brain-Computer Interface. *Computational Intelligence and Neuroscience*, Vol.2018, No.2018, 1-12, 2018.
- [10] H. Fauzi, M. I. Shapiai and U. Khairuddin. Transfer Learning of BCI Using CUR Algorithm. *Journal of Signal Processing Systems for Signal Image and Video Technology*, Vol.92, No.1, 109-121, 2020.
- [11] Rodrigues, C. Jutten, M. Congedo, et al. Riemannian Procrustes Analysis: Transfer Learning for Brain-Computer Interfaces. *IEEE Transactions on Biomedical Engineering*, Vol.66, No.8, 2390-2401, 2018.
- [12] F. Li, Y. Xia, F. Wang, et al. Transfer Learning Algorithm of P300-EEG Signal Based on XDAWN Spatial Filter and Riemannian Geometry Classifier. *Applied Sciences-Basel*, Vol.10, No.5, 2020.
- [13] P. Gaur, K. McCreddie, R. B. Pachori, et al. Tangent Space Features-Based Transfer Learning Classification Model for Two-Class Motor Imagery Brain-Computer Interface. *International Journal of Neural Systems*, Vol.29, No.10, 2019.
- [14] A. Salami, M. B. Khodabakhshi and M. H. Moradi. Fuzzy transfer learning approach for analysing imagery BCI tasks. *Artificial Intelligence & Signal Processing Conference*. Shiraz, IRAN, 2017.
- [15] A. M. Azab, Lyudmila Mihaylova, et al. Weighted Transfer Learning for Improving Motor Imagery-Based Brain-Computer Interface. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, Vol.27, No.7, 1352-1359, 2019.
- [16] M. Hajinoroozi, Z. Mao, Y. P. Lin, et al. Deep Transfer Learning for Cross-subject and Cross-experiment Prediction of Image Rapid Serial Visual Presentation Events from EEG Data. *Augmented Cognition: Neurocognition and Machine Learning*, Vol.10285, No.1, 45-55, 2017.
- [17] H. Wu, Y. Niu, F. Li, et al. A Parallel Multiscale Filter Bank Convolutional Neural Networks for Motor Imagery EEG Classification. *Frontiers in Neuroscience*, Vol.13, No.1275, 2019.
- [18] A. Farahat, C. Reichert, C. Sweeney-Reed, et al. Convolutional neural networks for decoding of covert attention focus and saliency maps for EEG feature visualization. *Journal of Neural Engineering*, Vol.16, No.10, 2019.
- [19] F. Fahimi, Z. Zhuo, W. B. Goh, et al. Inter-subject transfer learning with an end-to-end deep convolutional neural network for EEG-based BCI. *Journal of Neural Engineering*, Vol.16, No.2, 2019.
- [20] L. Xu, M. Xu, Y. Ke, et al. Cross-Dataset Variability Problem in EEG Decoding with Deep Learning. *Frontiers in Human Neuroscience*, 2020.
- [21] Z. RL, Q. Zong, and L. Q. Dou, et al. Hybrid deep neural network using transfer learning for EEG motor imagery decoding. *Biomedical Signal Processing and Control*, DOI: 10.1016/j.bspc.2020.102144, 2021.
- [22] X. Gao, S. Xiao, et al. A Deep Transfer Convolutional Neural Network Framework for EEG Signal Classification. *IEEE Access*, Vol.7, No.1, 112767-112776, 2019.
- [23] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. *Computer Science*, 2014.
- [24] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer and J. R. Wolpaw. BCI2000: a general-purpose brain-computer interface (BCI) system. *IEEE Transactions on Biomedical Engineering*, Vol.51, No.6, 1034-1043, 2004.
- [25] R. Leeb, C. Brunner, G. Müller-Putz, A. Schlögl and G. Pfurtscheller. BCI competition 2008-graz data set b. Graz University of Technology, Austria, 2008.